

Water Pollution: Computer Practice Dataset

ESE 2113 • Frequency distributions and exploratory analysis

Synthetic teaching data, not actual monitoring results. Each row represents one separate river sampling location measured during the same survey. BOD₅ is the primary variable for frequency analysis. DO is provided for scatter plots and correlation.

Sample ID	BOD ₅ (mg/L)	DO (mg/L)
W01	2	8.0
W02	3	8.0
W03	4	7.5
W04	5	7.2
W05	6	6.7
W06	7	7.6
W07	8	7.2
W08	9	7.1
W09	10	6.6
W10	11	7.2
W11	12	7.0
W12	13	6.5
W13	14	6.2
W14	15	6.1
W15	16	5.8
W16	17	6.0
W17	18	5.6
W18	19	5.9
W19	20	5.5
W20	21	5.3
W21	22	5.9
W22	24	4.8
W23	25	4.4
W24	27	4.8
W25	29	3.7
W26	30	4.4
W27	33	4.3
W28	36	3.5
W29	39	3.1
W30	44	1.8

Enter the headers in Excel cells A1:C1 and the observations in A2:C31. Keep each observation in its own row. Class intervals are practice choices, not regulatory pollution categories.

Air Pollution: Computer Practice Dataset

Synthetic teaching data. Each row represents one separate monitoring location, using illustrative 24-hour average concentrations from the same campaign. PM_{2.5} is the primary variable for frequency analysis. NO₂ supports scatter plots and correlation.

Sample ID	PM _{2.5} (µg/m ³)	NO ₂ (µg/m ³)
A01	12	14.2
A02	18	16.0
A03	21	12.5
A04	24	22.9
A05	27	14.6
A06	29	14.0
A07	30	14.0
A08	32	20.2
A09	35	22.3
A10	38	23.8
A11	40	22.2
A12	42	21.5
A13	44	27.2
A14	47	23.2
A15	49	23.0
A16	50	31.2
A17	51	27.9
A18	54	29.9
A19	58	31.9
A20	60	29.1
A21	63	30.3
A22	68	35.8
A23	70	32.7
A24	72	36.4
A25	78	37.4
A26	85	36.7
A27	89	44.7
A28	90	38.3
A29	96	45.2
A30	108	49.2

Use a separate Excel sheet. Enter headers in A1:C1 and observations in A2:C31. These datasets do not establish compliance or represent the university's research measurements.

Build the Frequency Tables in Excel

For WATER use [0,10), [10,20), [20,30), [30,40), [40,50). For AIR use [10,30), [30,50), [50,70), [70,90), [90,110). A bracket [includes the lower boundary; a parenthesis) excludes the upper boundary. Thus 30 belongs to [30,50), never to [10,30).

Create columns E:N with these headings: Lower, Upper, Width, Midpoint, Frequency, Relative f, Percentage, Cumulative f, Cumulative %, Frequency density. Enter the five lower/upper boundaries in E2:F6.

G2 (width): =F2-E2

H2 (midpoint): =(E2+F2)/2

I2 (frequency): =COUNTIFS(\$B\$2:\$B\$31,">="&E2,\$B\$2:\$B\$31,"<"&F2)

J2 (relative frequency): =I2/COUNT(\$B\$2:\$B\$31)

K2 (percentage): =100*J2

L2 (cumulative frequency): =SUM(\$I\$2:I2)

M2 (cumulative percentage): =100*L2/COUNT(\$B\$2:\$B\$31)

N2 (frequency density): =I2/G2

Fill each formula down to row 6. Checks: SUM(I2:I6)=30, SUM(J2:J6)=1, SUM(K2:K6)=100, L6=30 and M6=100. If using percentage formatting, format J directly as %. Do not multiply J by 100 and then apply % formatting.

Plots to produce

Histogram: use the stated class boundaries and touching bars. Equal-width bins may use frequency as height. A column chart with Gap Width = 0 is a teaching approximation for these equal-width bins; label the x-axis with the intervals.

Frequency polygon: use midpoint H as x and frequency I as y. Choose XY Scatter with straight lines and markers. Optional closure points: one class width before/after the first/last midpoint, with frequency 0.

Less-than ogive: use upper boundaries F as x and cumulative percentage M as y. Add the first lower boundary at 0%. Choose XY Scatter with straight lines and markers.

Scatter plot: use BOD₅ versus DO for water and PM_{2.5} versus NO₂ for air. Use an XY Scatter chart, not a line chart.

Analysis Tasks and Common Confusions

Tasks for both datasets

1. Calculate count, mean, median, minimum, maximum, range, sample standard deviation and coefficient of variation.
2. Find Q1, Q3 and IQR using QUARTILE.INC. State your quartile convention. Identify possible outliers using $Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$.
3. Identify the modal class. If several classes tie for the largest frequency, report all of them. A modal class differs from the raw-data mode.
4. Estimate the mean from the grouped table using midpoints. Compare it with the exact mean from the raw observations.
5. WATER: find the percentage with $BOD_5 < 20$ and with $BOD_5 \geq 30$. AIR: find the percentage with $PM_{2.5} < 70$ and with $PM_{2.5} \geq 90$. Read these results from the ogive.
6. Calculate Pearson correlation between the two pollutants/variables in each table. Explain the direction and strength. Correlation alone does not establish causation.
7. Compare distribution shape, spread and concentration of observations. Do not compare mg/L and $\mu g/m^3$ as if they measured the same quantity.

Excel formulas for raw-data analysis

`=AVERAGE(B2:B31)` `=MEDIAN(B2:B31)` `=COUNT(B2:B31)`
`=MIN(B2:B31)` `=MAX(B2:B31)` `=MAX(B2:B31)-MIN(B2:B31)`
`=STDEV.S(B2:B31)` $CV (\%) = 100 * STDEV.S(B2:B31) / AVERAGE(B2:B31)$
`=QUARTILE.INC(B2:B31,1)` `=QUARTILE.INC(B2:B31,3)`
`=CORREL(B2:B31,C2:C31)`
Grouped mean: `=SUMPRODUCT(H2:H6,I2:I6)/SUM(I2:I6)`

Meaning of each quantity

Frequency = number in a class. Relative frequency = fraction of the total. Cumulative frequency = running total below successive upper boundaries. Midpoint = a representative value for grouped calculations, not necessarily an observed value. Frequency density = count per unit of class width.

A histogram displays a quantitative variable across numeric intervals. A bar chart compares categories. An ogive never decreases. Grouped means and interpolated percentiles are approximations. Retain the raw values for exact statistics. All raw values here are distinct, so neither dataset has a raw-data mode.

Unequal Class Widths: The Essential Exercise

Regroup the WATER data into [0,10), [10,30), [30,50). Calculate the widths, frequencies and densities before plotting.

Class (mg/L)	Width	Frequency	Density f/width	Area = width $\times$ density
[0,10)	10	8	0.8	8
[10,30)	20	17	0.85	17
[30,50)	20	5	0.25	5

Use a numeric x-axis with boundaries 0, 10, 30 and 50. Bar widths must be 10, 20 and 20. Heights must be 0.80, 0.85 and 0.25. The first two bars have almost the same height, although their counts differ substantially. The wider interval contains more observations partly because it covers more of the measurement range.

A standard Excel category column chart gives every bar equal width, so it does not correctly display these unequal bins. In Excel, construct the rectangles as editable filled shapes on a numeric plotting grid: one x-axis unit must have the same physical spacing throughout. Choose a y-axis scale, then position each rectangle from its lower to upper boundary and set its height to the calculated density. Alternatively, use software with support for explicit unequal histogram breaks.

Verification: area of each bar = frequency. Total frequency-density area = 30. For probability density, divide each density by 30; heights become 0.02667, 0.02833 and 0.00833. Each area is the relative frequency, and total area = 1.

Teacher prompts

Ask: "Why does the [10,30) class contain more values but have nearly the same height as [0,10)?" Expected answer: it is twice as wide and has nearly the same count per unit width.

Ask: "If we used frequency as height, what would the areas be?" Answer: 80, 340 and 100. Their proportions would distort the true count proportions 8:17:5.

Ask: "Does a density height of 0.85 mean part of an observation?" Answer: no. It is an average count per measurement unit across the class.

Instructor Check Values

Water BOD₅

Class	Midpoint	f	Relative f	Cum. f	Cum. %
[0,10)	5.0	8	0.2667	8	26.67
[10,20)	15.0	10	0.3333	18	60.00
[20,30)	25.0	7	0.2333	25	83.33
[30,40)	35.0	4	0.1333	29	96.67
[40,50)	45.0	1	0.0333	30	100.00

n = 30; mean = 17.967; median = 16.500; min = 2; max = 44; range = 42; sample SD = 11.177; CV = 62.21%.

Q1 = 9.250; Q3 = 24.750; IQR = 15.500. Grouped mean = 18.333. No observations lie beyond the 1.5×IQR fences.

Air PM_{2.5}

Class	Midpoint	f	Relative f	Cum. f	Cum. %
[10,30)	20.0	6	0.2000	6	20.00
[30,50)	40.0	9	0.3000	15	50.00
[50,70)	60.0	7	0.2333	22	73.33
[70,90)	80.0	5	0.1667	27	90.00
[90,110)	100.0	3	0.1000	30	100.00

n = 30; mean = 52.667; median = 49.500; min = 12; max = 108; range = 96; sample SD = 25.060; CV = 47.58%.

Q1 = 32.750; Q3 = 69.500; IQR = 36.750. Grouped mean = 53.333. No observations lie beyond the 1.5×IQR fences.

Water: BOD₅ <20 = 18/30 = 60%; BOD₅ ≥30 = 5/30 = 16.67%. Pearson r (BOD₅, DO) = -0.977.

Air: PM_{2.5} <70 = 22/30 = 73.33%; PM_{2.5} ≥90 = 3/30 = 10%. Pearson r (PM_{2.5}, NO₂) = 0.961.

Water modal class: [10,20). Air modal class: [30,50). Correlations were built into these synthetic datasets for teaching and are not evidence of an actual environmental mechanism. Relative frequencies shown to four decimal places may have a small rounding discrepancy.